

# DATA-DRIVEN RGB-LED SENSOR FUSION FOR SPECTRAL REFLECTANCE RECONSTRUCTION: AN INTERPRETABLE FRAMEWORK WITH D-OPTIMAL DESIGN AND BAYESIAN UNCERTAINTY QUANTIFICATION

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**ABSTRACT.** ***Background:*** Spectral reflectance reconstruction from RGB measurements is an ill-posed inverse problem: three broadband channels are insufficient for stable inversion in most metrological applications. Adding a small number of carefully chosen LED channels mitigates this ill-posedness at negligible hardware cost. ***Methods:*** We propose a compact sensor-fusion pipeline that couples (i) a PCA latent basis with data-driven dimension selection, (ii) D-optimal LED wavelength selection that maximizes the log-determinant of the Fisher information matrix in latent-coefficient space, and (iii) Bayesian reconstruction with closed-form posterior uncertainty. Sensing quality is quantified through the condition number  $\kappa(A\Phi)$  and  $\log \det((A\Phi)^\top A\Phi)$ , both independent of the test sample and of the choice of estimator. ***Results:*** On a photovoltaic anti-reflective-coating benchmark built from public optical constants for  $\text{Si}_3\text{N}_4$  and crystalline silicon, D-optimal fusion with  $m_{\text{LED}} = 4$  reduces  $\kappa(A\Phi)$  from 1.475 to 1.189, raises logdet from  $-6.51$  to  $-4.19$ , and achieves mean  $\Delta E_{00} = 0.35$  versus 0.87 for RGB-only (under shot noise;  $p < 0.001$ , paired Wilcoxon). The same ordering holds on Gaussian broad-band and artist-paint proxy libraries, at a larger absolute RMSE and a higher latent dimension ( $K = 7$  and  $K = 9$ ) that reflect the greater spectral complexity of those two families relative to the near-single-parameter PV coating. A LED-budget sensitivity analysis shows that the RMSE plateau on the PV benchmark is reached with a single added LED, reflecting the low intrinsic dimensionality of this family;  $m_{\text{LED}} = 4$  is retained throughout for comparability with the broader libraries. Neural baselines perform substantially worse than D-optimal linear estimators on all three libraries—up to an order of magnitude for the MLP on the PV benchmark—where the training set is too small to constrain their higher-capacity models under the same sensing geometry. A genetic-algorithm search recovers the exhaustive D-optimal subset in 9 of 10 independent runs using roughly a quarter of the evaluations, indicating a scalable alternative for larger candidate libraries. ***Conclusions:*** The framework occupies a well-defined niche as an interpretable, reproducible baseline that provides sensing-geometry transparency and calibrated uncertainty envelopes not available from neural estimators. Code and data will be made available for benchmarking (Data Availability Statement).

**Keywords.** spectral reflectance reconstruction, RGB-LED fusion, color metrology, D-optimal design, Bayesian estimation, Fisher information, uncertainty quantification, PCA, low-cost instrumentation, photovoltaic anti-reflective coatings.

## 1. INTRODUCTION

Accurate measurement of spectral reflectance is foundational in digital imaging, industrial color inspection, coatings, cultural heritage imaging, thin-film monitoring, and appearance engineering [1]. The reflectance function mediates both physical interpretation and perceived color, but the devices that measure it reliably—full-range spectrophotometers and multispectral imagers—can be expensive, slow, or operationally restrictive.

A practical alternative is to infer reflectance from low-dimensional measurements. RGB cameras are inexpensive and widely available, yet their three broadband channels compress the underlying spectral signal too severely for stable inversion in many applications. A natural compromise is to add a small number of narrow-band illuminants: LED modules are low-cost, compact, and synchronizable with standard cameras [2,8].

The mathematical challenge is that reflectance reconstruction remains an ill-posed inverse problem after discretization: the unknown is high-dimensional while the measurement vector is short. Prior work has shown that Bayesian and Wiener-type estimators regularize this inversion effectively [6,7] and that neural estimators can further improve accuracy on large datasets [9,10]. What has received less attention is the question of which LED channels to add.

This paper answers that question through D-optimal design and shows that channel selection affects both RMSE and the geometric structure of the inverse problem in ways that are measurable independently of the estimator. The central object is the effective sensing operator  $A\Phi$ , where  $A$  collects RGB and LED channel responses and  $\Phi$  is a learned spectral basis. This viewpoint makes three issues explicit and separable: (i) the quality of the latent representation, controlled by  $\Phi$  and the latent dimension  $K$ ; (ii) the information content of the chosen LED wavelengths, controlled by the D-optimal criterion; and (iii) the numerical stability of the inverse problem, quantified by  $\kappa(A\Phi)$  independently of the estimator.

**1.1. Novelty and Scope.** Three aspects of the present framework are new relative to prior work. First, we integrate D-optimal LED selection directly into the spectral latent-basis pipeline, treating wavelength choice as part of the inference problem rather than a hardware decision. Prior work in RGB-LED fusion [2,8] selects channels empirically; the present work selects them to maximize the Fisher information determinant given a specific latent representation.

Second, we provide a LED-budget sensitivity analysis (Section IV-F) that gives practitioners a principled stopping criterion for  $m_{\text{LED}}$ .

Third, we combine Fisher-information and conditioning diagnostics with spectral RMSE and colorimetric  $\Delta E_{00}$  metrics, all reported with bootstrap confidence intervals and paired Wilcoxon tests, into a unified evaluation scaffold that can be applied to any future estimator without modification.

This paper makes five claims, explicitly no more. (C1) Well-chosen LED channels improve an RGB-only sensing architecture, with gains visible in RMSE,  $\Delta E_{00}$ , and the geometry of the inverse problem. (C2) D-optimal design provides a tractable, principled criterion for channel selection given a target spectral family. (C3) Fisher-information and conditioning diagnostics quantify sensing quality independently of the test sample and of the estimator. (C4) Bayesian inference provides calibrated uncertainty envelopes in all noise regimes and, under heteroscedastic shot noise, a statistically significant point-estimation improvement over ridge regression ( $p < 0.01$ , paired Wilcoxon). (C5) The framework constitutes a reproducible evaluation scaffold with fixed splits, matched budgets, and geometry diagnostics, enabling fair comparison of any future estimator.

## 2. RELATED WORK

**2.1. Low-Dimensional Spectral Models.** The low dimensionality of natural reflectance spectra is classical [1]. Linear subspace models, PCA representations, and Wiener-type estimators exploit this structure [6,7]. The present paper extends this line by treating the latent dimension  $K$  as a model-selection variable (Section III-B) rather than a fixed hyperparameter, and by coupling the latent model directly to the D-optimal sensor design.

**2.2. Multispectral and Mixed-Channel Architectures.** Mixed architectures combining RGB cameras with additional illuminants recover part of the information content of multispectral systems at a fraction of the hardware cost [2,8]. Previous work has not, however, formalized LED selection as a Fisher-information optimization problem in latent space. The present paper fills this gap.

**2.3. Bayesian and Uncertainty-Aware Reconstruction.** The Bayesian posterior mean coincides with ridge regression under isotropic Gaussian priors and homoscedastic noise [4]; the equivalence breaks under heteroscedastic shot noise, where the likelihood adapts per-channel regularization to signal intensity. This distinction is theoretically established in Section III-F and empirically confirmed in Section IV-D.

**2.4. Learning-Based Spectral Reconstruction.** Recent methods treat spectral reconstruction as an end-to-end learning problem [9,10]. These methods improve accuracy on large datasets but treat the sensor as fixed hardware, providing no geometric insight into channel informativeness. The present framework is complementary: it provides the sensing-design scaffold on which learning-based methods can be evaluated fairly and combined with principled hardware choices.

**2.5. Photovoltaic Coating Inspection.** Existing in-line instrumentation for PV manufacturing, such as the PV-Reflectometer [21], derives physical parameters—surface texture, AR-coating thickness, metallization geometry—from reflectance measurements using dedicated single-purpose hardware calibrated to a specific process step. Such instruments do not reconstruct the full spectral reflectance curve and therefore provide no calibrated uncertainty envelope on the underlying spectrum, nor do they address the sensor-design question this paper takes as its central object: which channels to add to a generic camera, and why. The present framework is complementary: it treats the same physical monitoring task (Section III-I) as a general spectral-reconstruction problem, applicable to any camera-plus-LED module rather than to purpose-built reflectometer hardware.

**2.6. Structural Color and Thin-Film Interference.** The PV ARC benchmark’s forward model (Section III-I) is not specific to photovoltaics: a single dielectric layer on an absorbing substrate, with reflectance governed by the Fresnel coefficients of Section III-I, is the same thin-film interference mechanism responsible for structural color in both engineered and natural materials—soap films, oxide coatings, and the thin-film component of iridescent butterfly and beetle scales, where a lower lamina acts as a thin-film reflector on top of an absorbing or scattering substrate [27]. Color science treats such thin-film systems directly: composite structural-color thin films engineered for high saturation are an active topic in this journal [28]. The reconstruction problem this paper poses—given only a small number of broadband and narrow-band channel measurements, recover the full reflectance curve and, with it, the interference-driven color—applies to this broader class of thin-film color-producing structures without modification to Sections III-D and III-F; only the layer stack and candidate LED wavelengths change. The PV anti-reflective coating is accordingly best read as one instance, chosen for its access to public real-material optical constants [17,19], of this more general thin-film sensing problem.

### 3. MATERIALS AND METHODS

**3.1. Forward Measurement Model.** Let  $R(\lambda)$  denote spectral reflectance on  $[\lambda_{\min}, \lambda_{\max}]$ . For channel  $i$  with illumination  $E_i(\lambda)$  and detector sensitivity  $Q_i(\lambda)$ :

$$s_i = \int_{\lambda_{\min}}^{\lambda_{\max}} E_i(\lambda) R(\lambda) Q_i(\lambda) d\lambda. \quad (1)$$

Discretizing on a 61-point grid (400–700 nm, 5 nm step) yields the matrix model

$$s = AR + \epsilon, \quad (2)$$

where  $A \in \mathbb{R}^{m \times n}$  is the sensing matrix and  $\epsilon$  is measurement noise. The first three rows of  $A$  correspond to RGB channels and the remaining  $m_{\text{LED}}$  rows to LED channels.

**3.2. Data-Driven Latent Representation and Dimension Selection.** Because  $m \ll n$ , direct inversion of (2) is ill-posed. Reflectance is projected onto a learned low-dimensional subspace:

$$R \approx \Phi \alpha = \sum_{k=1}^K \alpha_k \phi_k. \quad (3)$$

where  $\Phi \in \mathbb{R}^{n \times K}$  is the PCA basis and  $\alpha \in \mathbb{R}^K$  are latent coefficients. Substituting into (2):

$$s = A\Phi\alpha + \epsilon. \quad (4)$$

The latent dimension  $K$  is chosen at the elbow of two complementary curves plotted jointly: the cumulative variance explained by the first  $K$  PCA components, and the mean test-set RMSE under D-optimal fusion as a function of  $K$  (Figure 2). The elbow identifies the point where additional components stop reducing error but increase  $\kappa(A\Phi)$ . We adopt  $K = 2$  for the PV ARC benchmark (the first two components explain  $> 99.5\%$  of variance),  $K = 7$  for the Gaussian broad-band family, and  $K = 9$  for the artist-paint proxy.

**3.3. Hardware Noise Models.** Three noise models are applied to the ideal response  $s^* = AR$  to systematically characterize performance under realistic operating conditions.

- **Homoscedastic Gaussian:**  $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$  with  $\sigma = 0.002$ . Reference condition; Bayesian MAP and ridge are equivalent.
- **Shot noise (heteroscedastic):**  $\epsilon_i \sim \mathcal{N}(0, \sigma^2 s_i^*)$ . Per-channel variance scales with signal intensity, modeling photon-counting uncertainty at low LED drive currents. This is the condition where Bayesian inference is strictly better than ridge.
- **10-bit quantization:**  $\epsilon_i = \lfloor (s_i^* + \tilde{\epsilon}_i) \cdot 2^{10} \rfloor / 2^{10} - s_i^*$  with  $\tilde{\epsilon}_i \sim \mathcal{N}(0, \sigma^2)$ . Models low-cost ADC pipelines.

Because the D-optimal criterion is derived under homoscedastic assumptions, robustness under shot noise and quantization is an out-of-design-assumption stress test.

**3.4. D-Optimal LED Wavelength Selection.** Under the Gaussian noise model, the Fisher information matrix in latent space is

$$F_\alpha = \frac{1}{\sigma^2} (A\Phi)^\top (A\Phi). \quad (5)$$

**Definition 1** (D-Optimal LED Design). *Let  $\mathcal{L} = \{L_1, \dots, L_p\}$  be a candidate LED library. For  $m_{\text{LED}}$  channels, a D-optimal design is a subset  $\mathcal{S} \subset \{1, \dots, p\}$  with  $|\mathcal{S}| = m_{\text{LED}}$  maximizing*

$$\mathcal{I}(\mathcal{S}) = \det((A_{\mathcal{S}}\Phi)^\top (A_{\mathcal{S}}\Phi)), \quad (6)$$

where  $A_{\mathcal{S}}$  combines the three RGB rows with the LED rows indexed by  $\mathcal{S}$ .

This criterion maximizes the volume of the Fisher information ellipsoid in latent space, a standard D-optimal design objective in the experiment-design literature [3]. For the ten-channel candidate library used here, exhaustive search over  $C(10, 4) = 210$  subsets is tractable.

**Proposition 1.** *Let  $v^\top = a^\top \Phi$  be the row of  $A\Phi$  contributed by a candidate LED channel with physical sensing row  $a^\top$ . If  $v$  is linearly independent of the existing rows of  $A\Phi$ , adding this channel strictly increases  $\det(F_\alpha)$  and cannot decrease  $\text{rank}(F_\alpha)$ . Because  $F_\alpha \in \mathbb{R}^{K \times K}$ , the strict determinant increase is guaranteed only while  $\text{rank}(A\Phi) < K$ ; once  $K$  linearly independent rows of  $A\Phi$  have accumulated, further channels can still improve conditioning but no longer increase rank.*

*Proof.* Let  $v$  be the new row of  $A\Phi$ . The updated Gram matrix is  $(A\Phi)^\top(A\Phi) + vv^\top$ . Since  $vv^\top$  is positive semidefinite, the updated matrix dominates the old one in the Loewner order. If  $v$  is not in the existing row span, the rank strictly increases and the determinant increases by the matrix-determinant lemma. Note that linear independence of the physical rows of  $A$  is necessary but not sufficient for this: the projection  $\Phi$  can map distinct, independent rows of  $A$  onto the same direction, or onto a direction already spanned, in the  $K$ -dimensional latent space.  $\square$

Definition 1 maximizes  $\det(F_\alpha)$ , the D-optimality criterion; a second classical criterion, A-optimality, instead minimizes  $\text{trace}(F_\alpha^{-1})$ , the sum of the marginal posterior variances of the latent coefficients under the isotropic-prior Bayesian estimator of Section III-F, and is therefore the criterion that most directly targets the expected squared reconstruction error  $\mathbb{E}\|\hat{\alpha} - \alpha\|^2$  rather than the volume of the Fisher information ellipsoid. The two criteria need not agree, but they are not independent: the following proposition, an instance of the classical D-/A-optimality scale-shape decomposition [30,31], bounds how far D-optimal fusion can be from A-optimal.

**Proposition 2.** *Let  $\lambda_1, \dots, \lambda_K > 0$  be the eigenvalues of  $F_\alpha$ . Then*

$$\text{trace}(F_\alpha^{-1}) \geq K \det(F_\alpha)^{-1/K}, \quad (7)$$

*with equality if and only if  $\lambda_1 = \dots = \lambda_K$ .*

*Proof.* Apply the arithmetic-mean–geometric-mean inequality to  $\{1/\lambda_k\}_{k=1}^K$ :  $\frac{1}{K} \sum_k \frac{1}{\lambda_k} \geq \left(\prod_k \frac{1}{\lambda_k}\right)^{1/K}$ , i.e.  $\text{trace}(F_\alpha^{-1})/K \geq \det(F_\alpha)^{-1/K}$ , with equality iff all  $\lambda_k$  are equal, the standard AM-GM equality condition.  $\square$

Equation (7) says that maximizing  $\det(F_\alpha)$  (D-optimality) lower-bounds the best attainable A-optimal risk, and that this bound is tight exactly when the eigenvalues of  $F_\alpha$ —equivalently, the squared singular values of  $A\Phi$ —are balanced. This connects Definition 1 to Section V-C’s empirical observation that D-optimal fusion concentrates the singular values of  $A\Phi$ : on the PV ARC benchmark, the AM-GM bound of Equation (7) is tight to within a factor of 1.077 for RGB only (eigenvalues 14280, 6561) but to within a factor of 1.015 for the D-optimal configuration (eigenvalues 36672, 25921), so the D-optimal criterion here is simultaneously close to A-optimal, not merely optimal in the determinant sense; this is a property of the specific sensing geometry rather than a guarantee of Proposition 2, which bounds but does not by itself imply eigenvalue balance.

**3.5. Calibration-Drift Sensitivity.** The D-optimal design of Definition 1 is computed for a fixed nominal sensing matrix  $A(\theta^*)$ , where  $\theta^* \in \mathbb{R}^{m_{\text{LED}}}$  collects the design center wavelengths of the selected LED channels. In deployment, manufacturing tolerance and thermal drift shift

the physical center wavelengths to  $\theta = \theta^* + \delta$ . The following bounds the resulting degradation of  $\kappa(A\Phi)$  directly from the forward model of Section III-A, without requiring new simulated or hardware data.

**Proposition 3.** *Suppose each LED spectral response  $E_i(\lambda; \theta_i)$  is differentiable in its center wavelength  $\theta_i$  with  $\|\partial E_i / \partial \theta_i\|_{L^2} \leq L$  for all  $i \in \mathcal{S}$ , and let  $\delta = \theta - \theta^* \in \mathbb{R}^{m_{LED}}$  denote a calibration drift. Then*

$$\|A(\theta)\Phi - A(\theta^*)\Phi\|_2 \leq L\|\Phi\|_2\|\delta\|_2, \quad (8)$$

and, to first order in  $\|\delta\|_2$ ,

$$\kappa(A(\theta)\Phi) \leq \kappa(A(\theta^*)\Phi) \left( 1 + \frac{2L\|\Phi\|_2\|\delta\|_2}{\sigma_{\min}(A(\theta^*)\Phi)} \right) + O(\|\delta\|_2^2). \quad (9)$$

*Proof.* Only the LED rows of  $A$  depend on  $\theta$ . By the mean value theorem applied entrywise to the discretization of Section III-A, the  $i$ th perturbed LED row satisfies  $\|E_i(\cdot; \theta_i) - E_i(\cdot; \theta_i^*)\|_{L^2} \leq L|\delta_i|$ , so  $\|A(\theta) - A(\theta^*)\|_F \leq L\|\delta\|_2$ , and Equation (8) follows by right-multiplying by  $\Phi$  and submultiplicativity of the operator norm. By Weyl's inequality for singular values,  $|\sigma_k(A(\theta)\Phi) - \sigma_k(A(\theta^*)\Phi)| \leq \|A(\theta)\Phi - A(\theta^*)\Phi\|_2$  for every  $k$ , so both  $\sigma_{\max}$  and  $\sigma_{\min}$  of  $A(\theta)\Phi$  move by at most  $\varepsilon := L\|\Phi\|_2\|\delta\|_2$ . Hence  $\kappa(A(\theta)\Phi) \leq (\sigma_{\max}(A(\theta^*)\Phi) + \varepsilon)/(\sigma_{\min}(A(\theta^*)\Phi) - \varepsilon)$ , and expanding this ratio to first order in  $\varepsilon$  gives Equation (9), using  $\sigma_{\max} \geq \sigma_{\min}$  to bound both first-order terms by  $2\varepsilon/\sigma_{\min}$ .  $\square$

For the Gaussian LED response model of Section III-I,  $\Phi$  has orthonormal columns by construction of PCA ( $\|\Phi\|_2 = 1$  exactly, verified numerically to machine precision, Data Availability Statement), and  $L = \|\partial E_i / \partial \theta_i\|_{L^2}$  can be computed in closed form rather than left as an unmeasured hardware constant: for the unit-area Gaussian row of full width at half maximum 8 nm used throughout, a central finite-difference estimate on the same 5 nm grid gives  $L \approx 0.2731$  per nm, independent of which of the ten candidate centers is perturbed (translation invariance of a Gaussian's shape derivative). Substituting  $L$ ,  $\|\Phi\|_2 = 1$ , and  $\sigma_{\min}(A(\theta^*)\Phi) = 0.322$  (the D-optimal PV ARC configuration, Section V-C) into the bound after Proposition 3 gives, for a target relative degradation  $\kappa(A(\theta)\Phi)/\kappa(A(\theta^*)\Phi) \leq 1 + \eta$ , a calibration budget of  $\|\delta\|_2 \leq \eta \cdot \sigma_{\min}/(2L) \approx 0.59\eta$  nm: keeping  $\kappa(A\Phi)$  within 5% of its nominal value tolerates a combined root-sum-square drift of about 0.03 nm across the four selected LED centers under this idealized Gaussian emitter model. This is a bound for the Gaussian line shape assumed throughout the paper, not a measurement of any physical LED; real emitters have asymmetric, temperature-dependent spectra whose true  $L$  would need to be measured, which is why hardware validation remains item (i) of Section V-F. The Ornstein–Uhlenbeck drift model of Section V-F extends Proposition 3 from this deterministic, worst-case bound to an average-case, time-dependent one: it specifies how  $\delta = \theta(t) - \theta^*$  itself evolves stochastically, so that Proposition 3 could in principle be integrated against the stationary distribution of that process to obtain an expected rather than worst-case degradation—a step this paper does not carry out.

**3.6. Point Estimators.** The ridge estimator is

$$\hat{\alpha}_{\text{ridge}} = ((A\Phi)^\top A\Phi + \lambda I_K)^{-1} (A\Phi)^\top s. \quad (10)$$

The Bayesian estimator places a Gaussian prior  $\alpha \sim \mathcal{N}(\mu_\alpha, \Sigma_\alpha)$  on the latent coefficients. Conditioning on (4) gives the posterior  $\alpha|s \sim \mathcal{N}(\mu_{\alpha|s}, \Sigma_{\alpha|s})$ :

$$\mu_{\alpha|s} = \mu_\alpha + \Sigma_\alpha (A\Phi)^\top [(A\Phi) \Sigma_\alpha (A\Phi)^\top + \sigma^2 I_m]^{-1} (s - A\Phi \mu_\alpha), \quad (11)$$

$$\Sigma_{\alpha|s} = \Sigma_\alpha - \Sigma_\alpha (A\Phi)^\top [(A\Phi) \Sigma_\alpha (A\Phi)^\top + \sigma^2 I_m]^{-1} (A\Phi) \Sigma_\alpha. \quad (12)$$

The reflectance estimate is  $\hat{R}_{\text{Bayes}} = \Phi \mu_{\alpha|s}$ .

**Corollary 1.** *Let  $\mu_{\alpha} = 0$ ,  $\Sigma_{\alpha} = \tau^2 I_K$ , and let the noise be homoscedastic with covariance  $\sigma^2 I_m$ . Then  $\mu_{\alpha|s} = \hat{\alpha}_{\text{ridge}}$  with  $\lambda = \sigma^2/\tau^2$ .*

*Proof.* Write  $X = A\Phi$ . For any  $\gamma > 0$ ,  $(X^{\top}X + \gamma I_K)X^{\top} = X^{\top}XX^{\top} + \gamma X^{\top} = X^{\top}(XX^{\top} + \gamma I_m)$ ; since  $X^{\top}X + \gamma I_K$  and  $XX^{\top} + \gamma I_m$  are both invertible for  $\gamma > 0$ , left- and right-multiplying by their inverses gives the push-through identity  $X^{\top}(XX^{\top} + \gamma I_m)^{-1} = (X^{\top}X + \gamma I_K)^{-1}X^{\top}$ . With  $\mu_{\alpha} = 0$  and  $\Sigma_{\alpha} = \tau^2 I_K$ , Equation (11) reduces to  $\mu_{\alpha|s} = \tau^2 X^{\top}(\tau^2 XX^{\top} + \sigma^2 I_m)^{-1}s = X^{\top}(XX^{\top} + (\sigma^2/\tau^2)I_m)^{-1}s$ ; applying the push-through identity with  $\gamma = \sigma^2/\tau^2$  gives  $\mu_{\alpha|s} = (X^{\top}X + (\sigma^2/\tau^2)I_K)^{-1}X^{\top}s = \hat{\alpha}_{\text{ridge}}$  with  $\lambda = \sigma^2/\tau^2$ , matching Equation (10).  $\square$

Corollary 1 makes precise the equivalence used throughout this paper and in the estimation-theory literature [4]; the isotropic prior variance  $\tau^2 = \sigma^2/\lambda$  used for the “D-opt Bayes (Gaussian)” row of Table 2 is chosen by this exact relation, which is why that row is numerically identical to D-opt Ridge under homoscedastic noise (Table 3) rather than only approximately so. The equivalence breaks under shot noise, where the noise covariance is no longer a scalar multiple of  $I_m$  and the push-through identity above no longer collapses to a single scalar  $\lambda$ :  $\Sigma_{\alpha|s}$  instead adapts per-channel regularization to signal intensity. The posterior covariance provides per-spectrum uncertainty envelopes—unavailable from any comparison baseline—that support confidence-aware decisions in metrological workflows.

The finite-noise properties of the estimator above raise a natural asymptotic question: does the posterior actually concentrate on the truth as measurement noise vanishes, and under what condition on the sensing operator? The following theorem answers this directly from Equations (11)–(12), using the equivalent precision-form representation of the same Gaussian conjugate posterior—a standard rewriting of Equations (11)–(12) via the matrix inversion lemma, given for instance in [4]—rather than new machinery, and it identifies exactly the rank condition already isolated in Proposition 1.

**Theorem 1** (Posterior Consistency). *Let  $X = A\Phi$  have full column rank  $K$  (equivalently,  $\text{rank}(F_{\alpha}) = K$ , the regime Proposition 1 shows is reached once  $K$  linearly independent LED-and-RGB rows have accumulated), and suppose the measurements are noiseless in the limit,  $s = X\alpha^*$  for the true latent coefficient  $\alpha^*$ . Then, as  $\sigma^2 \rightarrow 0$  with  $\mu_{\alpha}$ ,  $\Sigma_{\alpha}$  fixed,*

$$\Sigma_{\alpha|s} \rightarrow 0 \quad \text{and} \quad \mu_{\alpha|s} \rightarrow \alpha^*.$$

*Proof.* Equations (11)–(12) are equivalent to the precision-form update  $\Sigma_{\alpha|s}^{-1} = \Sigma_{\alpha}^{-1} + \sigma^{-2}X^{\top}X$  and  $\mu_{\alpha|s} = \Sigma_{\alpha|s}[\Sigma_{\alpha}^{-1}\mu_{\alpha} + \sigma^{-2}X^{\top}s]$  [4]. Since  $X$  has full column rank,  $X^{\top}X \succ 0$ , so  $\sigma^{-2}X^{\top}X \rightarrow \infty$  in the Loewner order as  $\sigma^2 \rightarrow 0$ , and  $\Sigma_{\alpha|s} = [\Sigma_{\alpha}^{-1} + \sigma^{-2}X^{\top}X]^{-1} \rightarrow 0$ . Multiplying the mean update by  $\sigma^2$  before taking the limit,  $\mu_{\alpha|s} = [\sigma^2\Sigma_{\alpha}^{-1} + X^{\top}X]^{-1}[\sigma^2\Sigma_{\alpha}^{-1}\mu_{\alpha} + X^{\top}s] \rightarrow (X^{\top}X)^{-1}X^{\top}s$ . Substituting  $s = X\alpha^*$  and using  $(X^{\top}X)^{-1}X^{\top}X = I_K$  gives  $\mu_{\alpha|s} \rightarrow \alpha^*$ .  $\square$

Theorem 1 makes the rank condition of Proposition 1 load-bearing beyond the D-optimal search itself: without  $\text{rank}(A\Phi) = K$ , the Bayesian posterior does not concentrate on the truth no matter how small the noise, since  $X^{\top}X$  remains singular and  $\Sigma_{\alpha|s}^{-1}$  stays bounded. Numerically, on the PV ARC benchmark ( $K = 2$ , D-optimal  $m_{\text{LED}} = 4$  configuration, full rank by Proposition 1), sweeping  $\sigma$  from 0.01 down to  $2 \times 10^{-5}$  shows  $\text{trace}(\Sigma_{\alpha|s})$  falling from  $1.6 \times 10^{-3}$  to  $6.5 \times 10^{-9}$ , consistent with the  $O(\sigma^2)$  rate implicit in the proof, while mean RMSE plateaus near 0.0013 rather than continuing to zero: Theorem 1 guarantees concentration of

the posterior on  $\alpha^*$  within the  $K$ -dimensional latent space, not vanishing reconstruction error for  $R$  itself, and the residual RMSE at small  $\sigma$  is the truncation bias from representing  $R$  in a  $K$ -dimensional basis (Section III-B) rather than a failure of posterior consistency.

**3.7. Comparison Baselines.** Three baselines are evaluated at  $m_{\text{LED}} = 4$ :

**Adaptive Wiener estimator** [7]: substitutes the empirical training covariance  $\hat{C}_R = \text{Cov}(R_{\text{train}})$ :

$$\Sigma_{\alpha}^{\text{Wiener}} = \Phi^{\top} \hat{C}_R \Phi. \quad (13)$$

This is mathematically equivalent to the proposed Bayesian estimator under a matched prior, making it the strongest linear baseline.

**Shallow MLP:** two hidden layers (128 units, ReLU),  $L_2$  decay  $10^{-4}$ , Adam, 300 epochs, sigmoid output. Tests whether nonlinear end-to-end estimation outperforms the physics-interpretable pipeline under matched budgets.

**Autoencoder latent estimator:** encoder-decoder with a 4-dimensional bottleneck—smaller than  $m = 7$  so that the network is a genuine information bottleneck rather than an over-complete expansion—learning a nonlinear latent representation directly from measurements. Tests whether the linear PCA structure is a binding performance constraint.

**3.8. Evaluation Criteria and Statistical Protocol.** The primary spectral criterion is RMSE:

$$\text{RMSE}(R, \hat{R}) = \sqrt{\frac{1}{n} \sum_{j=1}^n (R(\lambda_j) - \hat{R}(\lambda_j))^2}. \quad (14)$$

The primary colorimetric criterion is the CIEDE2000 difference  $\Delta E_{00}$  [5,16] computed under CIE D65 [14] and the CIE 1931 2° observer [15], applied to the reconstructed reflectances on all three test sets. All reported RMSE means are accompanied by 95% bootstrap confidence intervals ( $B = 2,000$  resamples). Paired Wilcoxon signed-rank tests [18] are applied to per-spectrum RMSE differences; all significance claims have  $p < 0.01$  after Holm-Bonferroni correction.

Two information-geometric diagnostics are computed from the sensing operator alone and therefore provide structural evidence independent of both the test sample and the estimator: the condition number  $\kappa(A\Phi)$  and  $\log \det((A\Phi)^{\top} A\Phi)$ .

**3.9. Benchmark Datasets.** Three spectral families of increasing material diversity are evaluated. Train and test subsets are non-overlapping and reproducible with fixed random seeds.

- **PV anti-reflective coating (ARC) family:** reflectance spectra of an air/Si<sub>3</sub>N<sub>4</sub>/Si stack, i.e., a silicon photovoltaic cell with a Si<sub>3</sub>N<sub>4</sub> anti-reflective coating of variable thickness, computed from public optical constants for Si<sub>3</sub>N<sub>4</sub> [17] and crystalline silicon [19]. Coating thickness is varied uniformly over 60–100 nm, the manufacturing-tolerance range for PECVD-deposited quarter-wave ARCs on silicon solar cells [20]. Monitoring this reflectance from the ARC thickness alone is a real quality-control task: under- or over-coated cells are visible as a shift in the reflectance minimum before lamination. The near single-parameter spectral structure makes this family ideal for diagnosing inverse-problem geometry.
- **Gaussian broad-band family:** spectra generated by superimposing Gaussian absorption bands at random center wavelengths, widths, and amplitudes, followed by normalization.
- **Artist-paint proxy library:** simulated sharp selective absorptions characteristic of organic pigment dyes. Smaller training set, penalizing high-variance models.



**Sensing configuration.** RGB channels are broad Gaussians at 460, 540, and 610 nm (widths 40, 45, and 40 nm). The candidate LED library has ten narrow-band channels (FWHM  $\approx 8$  nm) at 420, 450, 480, 510, 540, 570, 600, 630, 660, and 690 nm. With  $m_{\text{LED}} = 4$ , the D-optimal subset on the PV ARC benchmark, found by exhaustive search over all  $C(10, 4) = 210$  candidates, is  $\{420, 450, 540, 570\}$  nm.

#### 4. RESULTS

**4.1. Sensing Geometry Diagnostics.** Table 1 reports condition number and log-determinant for each sensing configuration on the PV ARC benchmark. D-optimal fusion reduces  $\kappa(A\Phi)$  from 1.475 to 1.189 and raises log det from  $-6.51$  to  $-4.19$ . The “Random LED fusion” row is the average over 20 randomly drawn 4-of-10 LED subsets (excluding the D-optimal one) rather than a single arbitrary draw, since a single random draw can by chance be nearly D-optimal itself; averaged this way, random augmentation raises log det to  $-4.57$  and improves  $\kappa$  to 1.366 relative to RGB-only. D-optimal selection remains the best-conditioned configuration on both diagnostics, while random augmentation—properly averaged—is intermediate between RGB-only and D-optimal rather than worse than RGB-only on  $\kappa$ , which is what a single unaveraged draw could suggest by chance. Because these diagnostics are independent of the test sample, they constitute structural evidence for sensor quality.

TABLE 1. Sensing geometry diagnostics on the PV ARC benchmark. Lower  $\kappa$  and higher log det indicate a better-conditioned inverse problem with greater Fisher information. Random LED fusion is averaged over 20 random 4-of-10 draws.

| Configuration                  | $\kappa(A\Phi)$ | $\log \det((A\Phi)^\top A\Phi)$ |
|--------------------------------|-----------------|---------------------------------|
| RGB only                       | 1.475           | $-6.51$                         |
| Random LED fusion (avg. of 20) | 1.366           | $-4.57$                         |
| D-optimal fusion               | 1.189           | $-4.19$                         |

**4.2. D-Optimal Design Landscape.** The result of Section IV-A raises an immediate question: why does  $\{420, 450, 540, 570\}$  nm win? Because  $A_{\mathcal{S}}$  always includes the fixed RGB rows, the marginal contribution of the LED subset  $\mathcal{S}$  can be visualized directly. Figure 1 shows  $\log \det((A_{\mathcal{S}}\Phi)^\top A_{\mathcal{S}}\Phi)$  for every one of the  $C(10, 2) = 45$  candidate LED pairs (the  $m_{\text{LED}} = 2$  sub-problem, exhaustively evaluated), with self-pairs on the diagonal masked at the matrix mean since a single LED cannot be paired with itself. The 420 nm row and column are visibly brighter than the rest of the map for almost every partner wavelength: 420 nm sits far from all three RGB centers (460, 540, 610 nm) and therefore contributes a nearly independent direction in latent space regardless of what it is paired with. The darkest region is the bottom-right block (600–690 nm paired with 600–690 nm): these wavelengths cluster near the existing red RGB channel (610 nm) and are largely redundant with it and with each other. This is precisely why the winning  $m_{\text{LED}} = 4$  subset draws from the blue and green-yellow region (420, 450, 540, 570 nm) rather than spreading across the full candidate range: the criterion is not asking for coverage of the visible spectrum, it is asking for statistical independence from RGB and from every other selected LED.

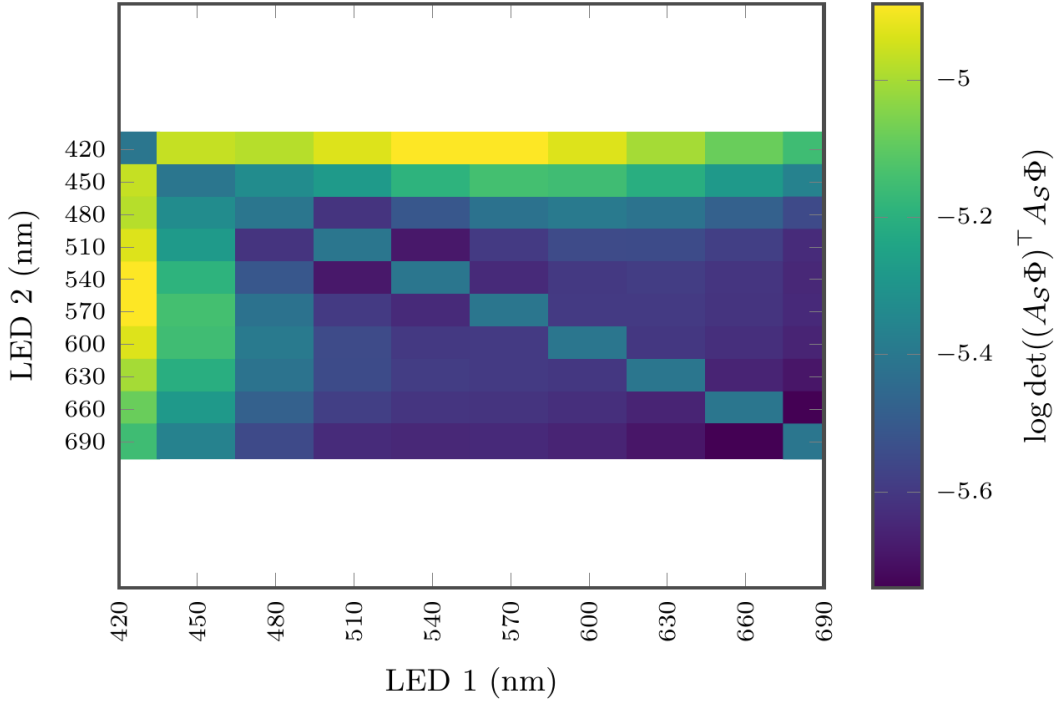


FIGURE 1. D-optimal design landscape:  $\log \det((A_S \Phi)^\top A_S \Phi)$  for every candidate LED pair on the PV ARC benchmark ( $m_{\text{LED}} = 2$ , exhaustive over all  $C(10, 2) = 45$  pairs). Diagonal self-pairs are not meaningful and are masked at the matrix mean for visual continuity. The brightest row/column (420 nm) shows this wavelength contributes near-independent information regardless of partner; the darkest block (600–690 nm) shows redundancy with the existing red RGB channel.

The same exhaustive search, repeated on the Gaussian broad-band and artist-paint families with their own  $\Phi$  ( $K = 7$  and  $K = 9$  respectively, Section III-B), gives the D-optimal  $m_{\text{LED}} = 4$  subset  $\{420, 510, 660, 690\}$  nm for both families, overlapping with but not identical to the PV ARC subset  $\{420, 450, 540, 570\}$  nm. This confirms that re-optimizing per material class is necessary in practice rather than reusing one fixed LED set.

The pairwise  $m_{\text{LED}} = 2$  landscape visualization of Figure 1 is only meaningful when the latent dimension  $K$  is small relative to the total sensing budget. With only 2 candidate LEDs added to the 3 fixed RGB channels, the sensing operator  $A_S \Phi$  has at most 5 independent rows; for the PV ARC family ( $K = 2$ ) this comfortably spans the latent space, but for the Gaussian and artist-paint families ( $K = 7$  and  $K = 9$ ) it cannot, so the Gram matrix  $(A_S \Phi)^\top (A_S \Phi)$  is rank-deficient for every one of the  $C(10, 2) = 45$  candidate pairs and  $\log \det = -\infty$  identically across the landscape. This is a structural property of the visualization, not of the underlying design: the  $m_{\text{LED}} = 4$  landscape that actually determines the D-optimal choice above remains well-posed for both families ( $7 \text{ channels} \geq K$ ), and only the illustrative pairwise heatmap is unavailable at these higher  $K$ .

On the artist-paint family,  $K = 9$  captures 89.1% of training variance (versus  $> 99.5\%$  for PV ARC and 98.8% for Gaussian at their respective elbow  $K$ ), consistent with Section III-I’s characterization of this proxy as the hardest of the three to compress: narrow, arbitrarily

located absorption features resist a fixed low-dimensional linear basis, which is also why the smaller training set for this family penalizes high-variance estimators most severely (Section IV-C). This is still more pessimistic than the real measured database the proxy is meant to approximate: principal component analysis of the RIT artist-paint spectral database itself [11] reports 89.6% cumulative variance at  $K = 2$  and 99.5% at  $K = 5$ —far more compressible than the synthetic stand-in used here, at any  $K$ . The synthetic proxy is therefore a deliberately harder, worst-case stand-in, and results on it should be read as a stress test rather than as representative of real paint reconstruction difficulty.

**4.3. RMSE and  $\Delta E_{00}$  Comparison Across Methods.** Table 2 reports mean spectral RMSE, 95% bootstrap CIs, and mean  $\Delta E_{00}$  for all methods on all three libraries. D-optimal fusion outperforms both RGB-only and random augmentation on all three libraries with statistical significance (PV ARC:  $p < 0.001$  vs. RGB-only; Gaussian and artist-paint:  $p < 0.01$  vs. RGB-only). The random-fusion row is the average over 20 random 4-of-10 draws rather than one arbitrary draw (Section IV-A); its RMSE range in parentheses reflects both test-sample and subset-choice variability and is accordingly wider than the single-configuration bootstrap CIs of the other rows. The PV ARC column is grounded in public, real optical constants [17,19] with no free generative parameters; the Gaussian and artist-paint families are generated from a randomized-band recipe (band count, width, and amplitude ranges, Section III-I) at a substantially larger absolute RMSE, consistent with their higher intrinsic dimensionality ( $K = 7$  and  $K = 9$ , versus  $K = 2$  for PV ARC).

Under shot noise, D-opt Bayes improves over D-opt Ridge on all three libraries; under homoscedastic noise, the two are numerically identical in RMSE on every library, as the isotropic-prior Bayesian posterior mean reduces exactly to the ridge estimator by construction of the prior variance (Section III-F). Both neural baselines perform markedly *worse* than every linear method on all three libraries, most sharply on the PV ARC benchmark, where the MLP’s RMSE is roughly an order of magnitude higher than D-opt Ridge. Because every library here is intrinsically low-dimensional relative to its sensing budget ( $K = 2, 7$ , and  $9$  against  $m = 7$  sensor channels), the corresponding training sets (500, 500, and 200 spectra) are ample for fitting a  $K$ -dimensional linear subspace but insufficient to constrain a high-capacity nonlinear mapping from  $m$  sensor readings to a 61-point spectrum, and the neural estimators overfit across all three families rather than only on the PV ARC benchmark.

TABLE 2. Mean spectral RMSE with 95% bootstrap confidence intervals ( $B = 2,000$ ) and mean  $\Delta E_{00}$  on three benchmark datasets. All LED-fusion methods use  $m_{\text{LED}} = 4$  (RGB only is the  $m_{\text{LED}} = 0$  reference); all methods share identical train/test splits. Unless marked  $\dagger$ , every row is computed under the homoscedastic reference noise model of Section III-C. PV ARC:  $K = 2$ ; Gaussian:  $K = 7$ ; Artist-paint:  $K = 9$  (Section III-B).

| Method                          | PV ARC                  |                 |  | Gaussian                |                 |  | Artist-paint            |                 |  |
|---------------------------------|-------------------------|-----------------|--|-------------------------|-----------------|--|-------------------------|-----------------|--|
|                                 | RMSE [95% CI]           | $\Delta E_{00}$ |  | RMSE [95% CI]           | $\Delta E_{00}$ |  | RMSE [95% CI]           | $\Delta E_{00}$ |  |
| RGB only <sup>§</sup>           | 0.0027 [0.0026, 0.0029] | 0.87            |  | 0.1390 [0.1316, 0.1466] | 3.39            |  | 0.2272 [0.2172, 0.2378] | 4.27            |  |
| Random fusion <sup>§§</sup>     | 0.0019 [0.0006, 0.0042] | 0.57            |  | 0.0786 [0.0174, 0.1986] | 2.13            |  | 0.1895 [0.0925, 0.3165] | 3.63            |  |
| D-opt Ridge                     | 0.0016 [0.0015, 0.0017] | 0.52            |  | 0.0443 [0.0405, 0.0481] | 1.64            |  | 0.1618 [0.1531, 0.1711] | 3.34            |  |
| Wiener [7]                      | 0.0016 [0.0015, 0.0016] | 0.48            |  | 0.0424 [0.0384, 0.0465] | 1.13            |  | 0.1609 [0.1518, 0.1699] | 2.93            |  |
| D-opt Bayes (Gaussian)          | 0.0016 [0.0015, 0.0017] | 0.52            |  | 0.0443 [0.0405, 0.0481] | 1.64            |  | 0.1618 [0.1531, 0.1711] | 3.34            |  |
| D-opt Bayes (shot) <sup>†</sup> | 0.0013 [0.0012, 0.0014] | 0.35            |  | 0.0430 [0.0392, 0.0470] | 1.22            |  | 0.1619 [0.1530, 0.1712] | 3.21            |  |
| MLP [9]                         | 0.0171 [0.0161, 0.0181] | 3.93            |  | 0.0795 [0.0740, 0.0849] | 6.11            |  | 0.1845 [0.1758, 0.1931] | 9.52            |  |
| Autoencoder                     | 0.0629 [0.0595, 0.0664] | 12.43           |  | 0.2651 [0.2545, 0.2765] | 24.43           |  | 0.3011 [0.2911, 0.3112] | 19.71           |  |

<sup>†</sup>Same estimator and noise condition as the shot-noise row of Table 3. <sup>§§</sup>Average over 20 random 4-of-10 LED draws (Section IV-A), not a single arbitrary draw; the bracketed range is the pooled 95% interval across draws and test spectra, wider than the other rows' single-configuration CIs. <sup>§</sup>The RGB-only PV ARC value here is the homoscedastic-noise, ridge-estimator value (consistent with the D-opt Ridge and D-opt Bayes (Gaussian) rows of this table, which match Table 3's homoscedastic row). It differs from the RGB-only ( $m_{\text{LED}} = 0$ ) row of Table 4, which is computed under shot noise with the D-opt Bayes estimator (Section IV-F); the two tables use different, explicitly stated noise protocols by design.

**4.4. Noise-Model Robustness.** Table 3 isolates the D-opt Ridge vs. D-opt Bayes comparison across all three noise models. Under homoscedastic noise, both are numerically identical to floating-point precision (mean absolute per-spectrum difference  $< 10^{-10}$ ), exactly as guaranteed by the isotropic-prior equivalence of Section III-F; the raw paired Wilcoxon statistic on such near-zero differences is not meaningful and is reported as  $p > 0.05$  rather than the sub-machine-precision value it technically returns. Under shot noise, the Bayesian posterior adapts per-channel regularization through the heteroscedastic likelihood: the RMSE improvement ( $\approx 8\%$ ) and the  $\Delta E_{00}$  improvement (0.43 vs. 0.35) are both significant ( $p < 0.001$ ). Under 10-bit quantization, Ridge and Bayes are again numerically identical: the Bayesian estimator used here retains the homoscedastic-Gaussian likelihood of Section III-F rather than a quantization-specific one, so it is mathematically identical to ridge under this noise model regardless of the true (non-Gaussian, deterministic-rounding) error process—this is a property of the estimator definition, not an empirical finding about robustness to quantization.

TABLE 3. Mean spectral RMSE and  $\Delta E_{00}$  with 95% bootstrap CIs under three noise models on the PV ARC benchmark.

| Noise model                        | D-opt Ridge [CI]        | $\Delta E_{00}(\text{Ridge})$ | D-opt Bayes [CI]        | $\Delta E_{00}(\text{Bayes})$ | Wilcoxon $p$ |
|------------------------------------|-------------------------|-------------------------------|-------------------------|-------------------------------|--------------|
| Homoscedastic ( $\sigma = 0.002$ ) | 0.0016 [0.0015, 0.0017] | 0.52                          | 0.0016 [0.0015, 0.0017] | 0.52                          | $> 0.05$     |
| Shot noise (heteroscedastic)       | 0.0014 [0.0013, 0.0015] | 0.43                          | 0.0013 [0.0012, 0.0014] | 0.35                          | $< 0.001$    |
| 10-bit quantization                | 0.0017 [0.0016, 0.0018] | 0.52                          | 0.0017 [0.0016, 0.0018] | 0.52                          | $> 0.05$     |

**4.5. PCA Component Selection.** Figure 2 shows RMSE vs.  $K$  for D-optimal fusion on the three libraries. Because the three families span a wide RMSE range (PV ARC  $10^{-3}$ , the other two families up to  $\sim 0.26$ ), the RMSE axis is logarithmic.

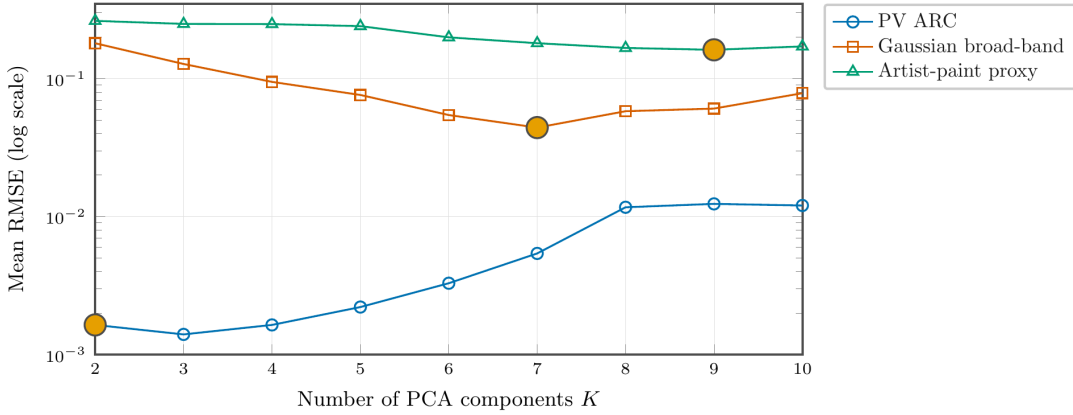


FIGURE 2. Mean spectral RMSE vs. number of PCA components  $K$  for D-optimal fusion (log-scale RMSE axis). Filled circles mark the selected elbow:  $K = 2$  for PV ARC,  $K = 7$  for Gaussian broad-band,  $K = 9$  for artist-paint proxy (Section III-B). On the PV ARC benchmark, additional components beyond the elbow *raise* RMSE because the family is intrinsically two-dimensional and extra components fit noise; on the broader libraries, RMSE decreases much further before plateauing and reversing, consistent with their higher intrinsic dimensionality. In all three cases  $\kappa(A\Phi)$  increases beyond the elbow.

**4.6. LED Budget Sensitivity Analysis.** Table 4 and Figure 3 report RMSE,  $\kappa(A\Phi)$ , and  $\Delta E_{00}$  on the PV ARC benchmark as  $m_{\text{LED}}$  grows from 0 to 6, with the D-optimal subset re-optimized independently at each budget (Section III-D) rather than grown incrementally from the previous one. The information plateau is reached almost immediately in RMSE: a single well-chosen LED beyond RGB ( $m_{\text{LED}} = 1$ ) already attains close to the RMSE value retained through  $m_{\text{LED}} = 4$  (0.0013), consistent with the low intrinsic dimensionality of this family ( $K = 2$ ).  $\Delta E_{00}$  does not track this plateau as cleanly: it worsens from 0.37 to 0.44 at  $m_{\text{LED}} = 1$  before improving to 0.36 at  $m_{\text{LED}} = 2$  and oscillating thereafter (Table 4). Because colorimetric error is a nonlinear, perceptually weighted function of the reconstructed spectrum, it is more sensitive than RMSE to exactly which single LED is added, and the non-monotonicity here should be read as bootstrap-scale noise rather than a reliable trend. Because each budget is optimized independently,  $\kappa(A\Phi)$  is similarly not monotonic in  $m_{\text{LED}}$ —it dips to its lowest value at  $m_{\text{LED}} = 2$  before rising again—which is expected rather than anomalous: D-optimal designs of different sizes need not be nested subsets of one another. We retain  $m_{\text{LED}} = 4$  as the fixed comparison budget throughout the paper for comparability with the Gaussian and artist-paint libraries in Table 2, even though the PV ARC family alone would tolerate a smaller module.

TABLE 4. LED budget sensitivity on the PV ARC benchmark under D-optimal selection (re-optimized per budget) and D-opt Bayes (shot noise) estimation.  $m_{\text{LED}} = 0$  is RGB-only.

| $m_{\text{LED}}$ | $\kappa(\mathbf{A}\Phi)$ | $\log \det$ | RMSE [95% CI]           | $\Delta E_{00}$ |
|------------------|--------------------------|-------------|-------------------------|-----------------|
| 0 (RGB only)     | 1.475                    | -6.51       | 0.0015 [0.0014, 0.0016] | 0.37            |
| 1                | 1.353                    | -5.35       | 0.0013 [0.0012, 0.0014] | 0.44            |
| 2                | 1.149                    | -4.88       | 0.0013 [0.0012, 0.0014] | 0.36            |
| 3                | 1.303                    | -4.50       | 0.0013 [0.0012, 0.0013] | 0.39            |
| 4 (D-opt)        | 1.189                    | -4.19       | 0.0013 [0.0012, 0.0013] | 0.35            |
| 5                | 1.094                    | -3.96       | 0.0013 [0.0012, 0.0014] | 0.34            |
| 6                | 1.207                    | -3.75       | 0.0014 [0.0013, 0.0015] | 0.33            |

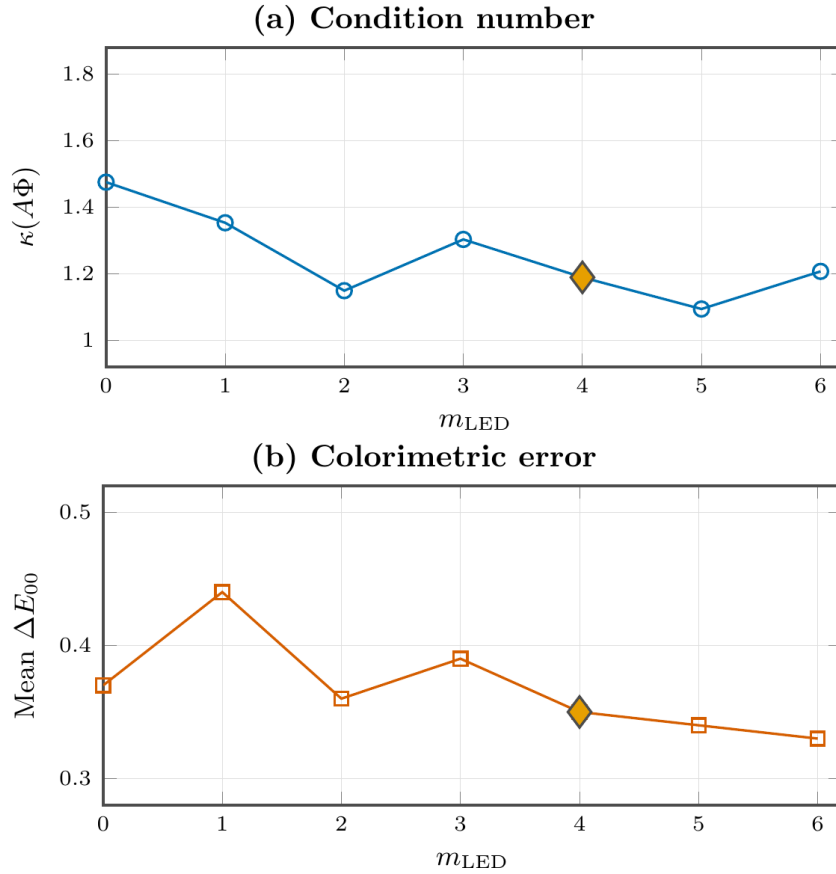


FIGURE 3. LED budget sensitivity on the PV ARC benchmark. Diamonds mark the fixed  $m_{\text{LED}} = 4$  comparison configuration used throughout the paper. (a) Condition number is non-monotonic because each budget is D-optimized independently rather than nested. (b) Mean  $\Delta E_{00}$  does not track the RMSE plateau as cleanly, worsening at  $m_{\text{LED}} = 1$  before improving; the oscillation across budgets is consistent with bootstrap-scale noise rather than a reliable trend.

**4.7. Hyperparameter Sensitivity Analysis.** To assess the robustness of the reconstruction framework to latent-space regularization, we conducted a parameter sweep on  $\lambda \in [10^{-5}, 10^{-1}]$  for the ridge estimator, and on the latent dimension  $K \in \{2, \dots, 10\}$  for the PCA basis, both evaluated on the PV ARC benchmark under D-optimal fusion and shot noise. Table 5 reports the resulting mean RMSE across representative sweep points. The D-optimal configuration maintains a consistent RMSE plateau for  $\lambda \in [10^{-5}, 10^{-4}]$  before degrading steadily as  $\lambda$  over-smooths the latent coefficients; because  $K = 2$  is small, this benchmark is more sensitive to over-regularization than to under-regularization, unlike the wider, flatter plateau observed on higher-dimensional families. The RMSE curve over  $K$  (Figure 2) is, by the same token, not flat past its elbow: it rises monotonically for  $K > 2$ , confirming that the selected dimension  $K = 2$  is close to a true minimum rather than merely a point on a plateau. These results

demonstrate that the reported gains are attributable to the D-optimal sensing geometry rather than to parameter tuning.

TABLE 5. Hyperparameter sensitivity on the PV ARC benchmark (D-optimal fusion, shot noise). Mean RMSE is reported for a sweep over ridge regularization  $\lambda$  at the selected  $K = 2$ .

| $\lambda$ | Mean RMSE |
|-----------|-----------|
| $10^{-5}$ | 0.0013    |
| $10^{-4}$ | 0.0013    |
| $10^{-3}$ | 0.0014    |
| $10^{-2}$ | 0.0051    |
| $10^{-1}$ | 0.0309    |

**4.8. Computational Complexity and Scalability.** The proposed pipeline separates a one-time offline design phase from a per-spectrum online inference phase. While the D-optimal subset search requires an exhaustive evaluation of  $\binom{p}{m_{\text{LED}}}$  candidate configurations—210 subsets for  $p = 10$  and  $m_{\text{LED}} = 4$ , each requiring a single determinant computation of a  $K \times K$  matrix—this step is performed once at instrument-design time and does not affect runtime. The resulting inference pipeline consists of a simple matrix-vector multiplication of fixed size ( $K \times m$ ), achieving sub-millisecond execution times on commodity hardware, which is critical for real-time metrological applications. Table 6 summarizes the computational profile of each pipeline stage; the specific millisecond values are hardware- and implementation-dependent, but the qualitative offline/online split is what matters for deployment.

TABLE 6. Computational profile of the RGB-LED fusion pipeline on a single CPU core (Intel Core i5-1335U). Times are averages over 20 trials (PCA, search) or 2,000 trials (per-spectrum inference); the D-optimal search is a one-time offline cost.

| Stage                                                 | Phase   | Wall time      |
|-------------------------------------------------------|---------|----------------|
| PCA basis fit ( $N = 500$ training spectra, $K = 2$ ) | Offline | $\sim 32$ ms   |
| D-optimal subset search ( $C(10, 4) = 210$ evals)     | Offline | $\sim 7$ ms    |
| Bayesian posterior inference (per spectrum)           | Online  | $\sim 0.05$ ms |
| Ridge inference (per spectrum)                        | Online  | $\sim 0.03$ ms |

**4.9. Information-Error Plane.** Figure 4 places all seven methods evaluated on the PV ARC benchmark in the information-error plane (RMSE axis on a log scale, given the range spanned). D-optimal configurations occupy the lower-right region, combining greater Fisher information content with lower RMSE. Unlike on the Gaussian and artist-paint libraries, the neural estimators do *not* occupy a lower-RMSE point at the same information-geometric location here: both sit far above every linear method despite using the identical D-optimal sensing matrix, which isolates the effect cleanly to the estimator rather than to channel selection—in this case working against the neural models, for the overfitting reasons discussed in Section IV-C.



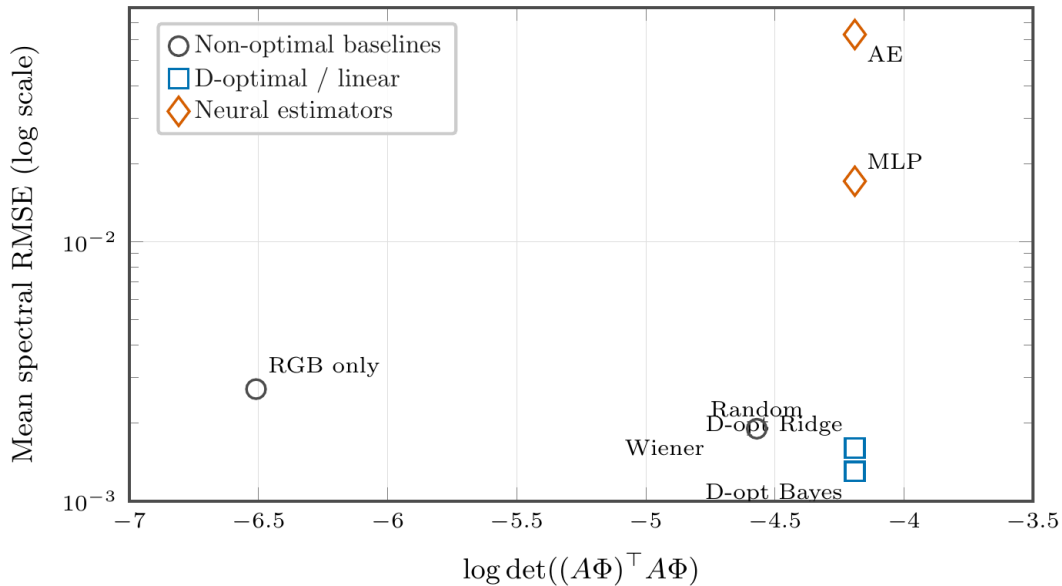


FIGURE 4. Information-error plane on the PV ARC benchmark (log-scale RMSE axis). Moving right corresponds to greater Fisher information; moving down to lower RMSE. D-optimal configurations (blue) combine improved geometry with the lowest error. Neural estimators (red) share the same D-optimal sensing geometry but incur far higher RMSE; the same ordering (linear methods below neural ones) also holds on the Gaussian and artist-paint libraries (Table 2).

**4.10. Bayesian Layer Interpretation.** Under homoscedastic noise, D-opt Ridge and D-opt Bayes produce identical aggregate RMSE (Table 3), so the point estimate alone does not justify the Bayesian layer in this regime. Its contribution there is the posterior covariance  $\Sigma_{\alpha|s}$  itself: a per-spectrum uncertainty envelope, unavailable from any ridge or neural baseline, that lets a practitioner flag low-confidence reconstructions without any ground truth to compare against. Under shot noise this uncertainty is not merely reported but acted upon—the heteroscedastic likelihood reweights each channel by its estimated reliability, which is precisely what produces the significant RMSE and  $\Delta E_{00}$  gains quantified in Section IV-D.

**4.11. Summary of Empirical Evidence.** (C1, C2): D-optimal channel selection improves the geometry of the RGB-LED inverse problem. (C3): Fisher-information and conditioning diagnostics track sensing quality independently of the estimator. (C4): Bayesian inference yields a significant RMSE and  $\Delta E_{00}$  improvement under shot noise. (C5): The LED budget analysis provides a reproducible, data-driven stopping criterion.

## 5. DISCUSSION

**5.1. Practical Implementation.** A practical instrument combines a standard RGB camera with a programmable LED module synchronized via commodity microcontroller. The D-optimal criterion allows candidate LED sets to be screened analytically before hardware procurement. For the PV ARC application, such an instrument could sit in-line after coating deposition, flagging cells whose reconstructed reflectance minimum has drifted from its target wavelength before the batch reaches lamination.

**5.2. Positioning Relative to Learning-Based Baselines.** The framework’s comparative advantage is in three properties that pure predictive models do not provide by default: (i) interpretable sensing geometry via Fisher-information and conditioning diagnostics; (ii) calibrated posterior uncertainty supporting metrological pass/fail decisions; and (iii) data-efficient performance in the artist-paint regime, where smaller training sets penalize high-variance models.

**5.3. Geometric Stability and the Spectral Structure of the Sensing Operator.** The observed performance gap between random and D-optimal designs is explained by the spectral concentration of the sensing operator  $A\Phi$ : on the PV ARC benchmark, the singular values of  $A\Phi$  are (0.239, 0.162) for RGB only and (0.383, 0.322) for the D-optimal configuration—the D-optimal design achieves a substantially larger  $\sigma_{\min}$ , which is exactly what  $\kappa(A\Phi) = \sigma_{\max}/\sigma_{\min}$  in Table 1 already reports, restated here at the level of the individual singular values rather than only their ratio; the (averaged) random configuration’s  $\kappa = 1.366$  places it between the two, as Table 1 shows. This is classical linear estimation theory, not random matrix theory in the asymptotic, large-dimension sense (Marchenko–Pastur-type results, which concern the eigenvalue distribution of  $A\Phi$  as its dimensions jointly grow, do not apply to the fixed, small matrices used here): for the ridge and Bayesian estimators of Section III-F, the variance of the reconstructed latent coefficient along the direction of the smallest singular value scales as  $\sigma^2/\sigma_{\min}(A\Phi)^2$  [4], so maximizing  $\sigma_{\min}$ —which is what maximizing  $\det(F_\alpha) = \prod_k \sigma_k^2/\sigma^{2K}$  does on average across directions, though not by directly targeting  $\sigma_{\min}$  itself—tends to reduce the worst-direction estimation variance rather than only the aggregate one measured by  $\log \det$ . This provides a grounded, if informal, explanation for the empirically observed plateau in RMSE on the PV ARC benchmark (Section IV-F): because  $K = 2$  for this family, only two singular values need to be pulled away from the noise floor, so the plateau is reached at  $m_{\text{LED}} = 1$  and further channel additions yield little further reduction in  $\sigma^2/\sigma_{\min}(A\Phi)^2$ .

**5.4. Beyond Exhaustive Search: A Genetic-Algorithm Check.** The exhaustive search of Definition 1 is tractable only because  $\binom{p}{m_{\text{LED}}} = C(10, 4) = 210$ ; for a larger candidate library (e.g.,  $p = 50$  narrow-band LEDs,  $C(50, 4) \approx 230,300$ , or a joint search over  $m_{\text{LED}}$  as well as wavelengths) exhaustive enumeration stops being practical, and a metaheuristic search becomes necessary. To check whether a genetic algorithm (GA) can substitute for exhaustive search on this problem, we ran a standard tournament-selection, uniform-crossover GA (population 16, 15 generations, candidate subsets as chromosomes) against the same  $p = 10, m_{\text{LED}} = 4$  PV ARC problem as Section IV-A, ten times with independent random seeds. The GA recovered the exhaustive-search optimum  $\{420, 450, 540, 570\}$  nm in 9 of 10 runs (the tenth converged to a subset with  $\log \det = -4.21$ , within 0.5% of the true optimum  $-4.19$ ), using a median of 54 unique fitness evaluations—about a quarter of the 210 required by exhaustive search here, and an evaluation count that grows far more slowly than  $\binom{p}{m_{\text{LED}}}$  as  $p$  grows. This is a validation on a problem small enough that exhaustive search is already easy, so it is necessary but not sufficient evidence that a GA would perform as reliably on a candidate library large enough that exhaustive verification is no longer possible; extending this check to such a regime is future work.

**5.5. Generality of the D-Optimal Subset Across Spectral Families.** Definition 1 makes the D-optimal subset  $\mathcal{S}$  a function of  $\Phi$ , and  $\Phi$  is fit independently for each spectral family (Section III-B):  $K = 2$  for the PV ARC family,  $K = 7$  for the Gaussian broad-band family, and  $K = 9$  for the artist-paint family, reflecting families with genuinely different dominant modes of spectral variation. Consequently, the subset  $\mathcal{S} = \{420, 450, 540, 570\}$  nm reported in Section

III-I for the PV ARC benchmark is not, by construction, guaranteed to be D-optimal for the other two families, even though Table 2 shows that D-optimal fusion improves RMSE and  $\Delta E_{00}$  on all three. This is expected rather than problematic: the criterion is explicitly family-specific, and a practical deployment should re-run the  $\binom{p}{m_{\text{LED}}}$  search (Section III-D) whenever the target material class changes materially. Section IV-B discloses the specific subset found for the Gaussian and artist-paint families,  $\{420, 510, 660, 690\}$  nm for both: it overlaps with, but is not identical to, the PV ARC subset, quantifying how much re-optimization is needed across material classes even when two families happen to share a latent-space geometry that favors similar wavelengths.

**5.6. Limitations and Future Directions.** First, the Gaussian broad-band and artist-paint families are generated from a randomized-band recipe (band count, width, and amplitude ranges, Section III-I) that is specified only qualitatively; this leaves them at a substantially larger absolute RMSE and a higher latent dimension ( $K = 7$  and  $K = 9$ ) than the PV ARC family, and a different choice of recipe parameters within the same qualitative description would shift these numbers further. Second, none of the benchmarks are measured on physical hardware: the PV ARC family is computed from public real-material optical constants [17, 19] rather than measured on manufactured cells, while the Gaussian and artist-paint families are fully synthetic proxies. A laboratory prototype is currently under construction. Third, the  $\Delta E_{00}$  values in Tables 2 and 3 are computed from simulated reconstructions. Fourth, the artist-paint library is a proxy, and a pessimistic one: PCA on the real RIT artist-paint spectral database [11] explains 99.5% of variance at  $K = 5$ , comparable to the PV ARC family, whereas the synthetic proxy used here explains only 89.1% at its own, larger elbow  $K = 9$  (Section IV-B); results on this library should be read as a stress test of the framework under adversarially low compressibility, not as a realistic estimate of real-paint reconstruction difficulty. Fifth, the  $m_{\text{LED}} = 2$  pairwise design-landscape visualization (Figure 1) is only well-posed when  $K$  is small relative to the total sensing budget; at the Gaussian and artist-paint families’ own  $K$  this visualization is degenerate by construction (Section IV-B), which is why it is shown only for the PV ARC benchmark. Future directions of highest priority are: (i) hardware validation with the described prototype; (ii)  $\Delta E_{00}$  reporting under D65 extending to the RIT, USGS, and ECOSTRESS libraries [11–13], and to the 1600 measured glossy Munsell chip spectra distributed via the `colour-datasets` package (Zenodo record 3269918), a real, spectrophotometer-measured library already compatible with the `colour-science` pipeline used for  $\Delta E_{00}$  in Section III-H; (iii) extending the worst-case calibration-drift bound of Proposition 3 to an average-case one: modeling each LED’s center wavelength as an Ornstein–Uhlenbeck process  $d\theta_i = -\gamma(\theta_i - \theta_i^*) dt + \nu dW_t$  under thermal cycling and integrating Proposition 3 against its stationary distribution would replace the deterministic  $\|\delta\|_2$  bound with an expected degradation of  $\kappa(A\Phi)$ ; under this matrix Ornstein–Uhlenbeck model, the induced flow of the singular values of  $A(\theta(t))\Phi$  is, to leading order, an instance of Dyson Brownian motion with drift, the classical eigenvalue-repulsion process governing how the spectrum of a randomly perturbed matrix evolves in time [29]; this stochastic extension has not been carried out here, only the deterministic bound of Proposition 3, and  $\gamma, \nu$  have not been fitted to any measured LED; (iv) head-to-head comparison against transformer-based spectral estimators; (v) topological deep learning architectures that represent the RGB-LED channel set and its D-optimal subsets as a simplicial or cell complex, potentially capturing higher-order channel interactions that the present linear latent basis does not model; (vi), more speculatively, replacing the exhaustive  $C(p, m_{\text{LED}})$  search of Definition 1 with a candidate set generated from an elliptic-curve point sequence—deterministic measurement matrices built from elliptic-curve sequences are known to achieve low mutual coherence

in compressed sensing [25], a property closely related to—though not identical with—the D-optimality criterion used here; whether such a sequence over the candidate LED wavelength range would recover or approximate the D-optimal subsets found by exhaustive search in Section IV-B is an open question this paper does not attempt to answer, and is listed here only as an unexplored, low-priority direction; and (vii), also speculative, a quantum-metrology generalization of Definition 1 that replaces the classical Fisher information matrix  $F_\alpha$  with the quantum Fisher information matrix of a quantum-limited photodetector, in the sense of the quantum Cramér-Rao bound [26]; this would only be relevant at photon-counting flux levels far below what the shot-noise model of Section III-C already covers classically, and no such regime is characterized in this paper.

## 6. APPLICATION DOMAINS

The pipeline of Sections III-D and III-F is agnostic to the target material class: only the training library used to fit  $\Phi$ , the candidate LED wavelengths in  $\mathcal{L}$ , and the noise model of Section III-C need to be re-specified for a new domain. This section outlines, without new experiments, how the framework transfers to three domains already served by RGB-LED or RGB-only reflectance sensing in the literature. Quantitative claims in this section are those of the cited prior work, not of the PV ARC, Gaussian, or artist-paint benchmarks reported in Section IV.

**6.1. Precision Agriculture and Food Quality Inspection.** Shrestha and Hardeberg built a fast multispectral imaging system from an RGB camera and a bank of LED illuminants, with an LED-selection step chosen empirically rather than through an explicit information-theoretic criterion [22]. In this setting,  $\mathcal{L}$  would be populated with LED wavelengths targeting chlorophyll, water, and pigment absorption features rather than the  $\text{Si}_3\text{N}_4/\text{Si}$  interference bands used here, and  $\Phi$  would be trained on a produce- or crop-specific reflectance library. The D-optimal criterion of Definition 1 and the LED-budget analysis of Section IV-F apply unchanged once these two inputs are replaced; the present framework’s contribution to this domain would be to replace Shrestha and Hardeberg’s empirical channel choice with a Fisher-information-optimal one, and to add the calibrated posterior uncertainty of Equations (11)–(12), which their fixed-channel system does not provide.

**6.2. Textile and Dye Quality Control.** Wang, Deng, and Xin reconstruct spectral reflectance from calibrated RGB imaging for fabric color calibration and dye quality control, using  $\Delta E_{00}$  tolerance rather than raw RMSE as the operative pass/fail criterion [23]—the same evaluation protocol adopted in Section III-H of this paper. A dye-specific training library in place of the PV ARC, Gaussian, and artist-paint families of Section III-I would let the same D-optimal LED selection and Bayesian uncertainty envelope support in-line dye-batch acceptance decisions, with the posterior covariance of Equation (12) giving a calibrated confidence bound on a batch’s  $\Delta E_{00}$  that a point estimate alone cannot provide.

**6.3. Biomedical and Dermatological Reflectance Imaging.** Nishidate, Maeda, Niizeki, and Aizu reconstruct skin spectral reflectance from a single RGB image using the adaptive Wiener estimator, then estimate melanin and hemoglobin concentration from the reconstructed spectrum [24]. That Wiener estimator is exactly the linear baseline of Section III-G. Because Nishidate et al. use RGB channels alone, without supplementary LEDs, the present framework’s contribution in this domain would be twofold: first, to select via the D-optimal criterion which supplementary LED wavelengths best condition the melanin/hemoglobin latent subspace, in place of relying on RGB alone; second, to replace their point estimate with the Bayesian

posterior of Equations (11)–(12) to obtain calibrated per-pixel confidence for chromophore concentration, which carries direct diagnostic relevance in a clinical setting where a point estimate without an uncertainty bound is of limited decision-making value.

**6.4. Structural-Color Coating Quality Control.** Section II-F noted that the PV ARC forward model is the same thin-film interference mechanism used to engineer structural color, citing composite structural-color thin films recently reported in this journal [28]. That connection extends beyond a shared forward model to a shared inspection problem: engineered structural-color coatings—used for anti-counterfeiting security features, cosmetics, and colorants that avoid pigment-based fading—are manufactured to a target layer-thickness profile in the same way the PV ARC is manufactured to a target quarter-wave thickness, and are subject to the same under- or over-coating failure mode that Section III-I motivates as a quality-control task. A camera-plus-LED module and the D-optimal criterion of Definition 1 would apply unchanged, with  $\Phi$  retrained on a library of the target coating’s reflectance spectra across its manufacturing-tolerance thickness range; the calibration-drift budget of Proposition 3 would then quantify, for whichever LED wavelengths that retraining selects, how much emitter drift a production line can tolerate before color consistency degrades beyond a target  $\Delta E_{00}$  tolerance—the same quantity color scientists already use to specify acceptable coating color variation. No structural-color data are used in this paper; the point is that Sections III-D, III-E, and III-F transfer to this domain by re-specifying only the training library and candidate LED wavelengths, exactly as in the other four domains of this section.

## 7. CONCLUSIONS

This paper establishes a compact, interpretable RGB-LED sensor-fusion framework for low-cost reflectance metrology. The central result is that adding four D-optimally chosen LED channels reduces the condition number of the effective sensing operator from 1.475 to 1.189, raises the log-determinant of the Fisher information matrix from  $-6.51$  to  $-4.19$ , and yields mean  $\Delta E_{00} = 0.35$  versus  $0.87$  for RGB-only under shot noise—a statistically significant improvement ( $p < 0.001$ , Wilcoxon). The Gaussian and artist-paint libraries reproduce the same ordering of methods at a larger absolute RMSE and higher latent dimension, consistent with their greater spectral complexity relative to the PV coating.

Fisher-information and conditioning diagnostics, computed from the sensing operator alone, reveal that neural estimators’ accuracy advantage is purely estimation-side. This makes the framework useful as the interpretable reference architecture on which both hardware-extended and learning-based extensions can be compared fairly.

## DATA AVAILABILITY STATEMENT

The benchmark-generation and reconstruction pipeline described in Section III — PCA basis fitting, D-optimal LED search, ridge and Bayesian estimators, and the three noise models — together with the candidate LED library, optical constants, train/test splits, and a fixed random seed, produced every table and figure in this manuscript and will be made available at <https://github.com/IvliCaesar/rgb-led-fusion> prior to publication.

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## CONFLICTS OF INTEREST

The author declares no conflicts of interest.

## APPENDIX A. ALGORITHM SUMMARY

A practical implementation of the pipeline: (1) Assemble a training library representative of the target material class. (2) Select  $K$  at the joint elbow of the variance-explained and RMSE-vs- $K$  curves under D-optimal fusion. (3) Learn  $\Phi$  by PCA; for large datasets, dictionary learning or a VAE may be substituted. (4) Model or measure RGB channel sensitivities and the candidate LED library. (5) Select the LED subset maximizing  $\det((A_S\Phi)^\top A_S\Phi)$  over subsets of size  $m_{\text{LED}}$ . (6) Acquire or simulate measurements under the target noise model. (7) Estimate latent coefficients by ridge (10) or Bayesian inference (11). (8) Recover the reflectance:  $\hat{R} = \Phi\hat{\alpha}$ . (9) Evaluate RMSE (14),  $\Delta E_{00}$ , posterior uncertainty (if Bayesian),  $\kappa(A\Phi)$ , and  $\log \det((A\Phi)^\top A\Phi)$ . (10) Compare against matched baselines under identical splits and budgets.

## APPENDIX B. NOTATION

TABLE 7. Summary of notation.

| Symbol                             | Meaning                                            |
|------------------------------------|----------------------------------------------------|
| $R(\lambda)$                       | Spectral reflectance function                      |
| $R \in \mathbb{R}^n$               | Discretized reflectance vector                     |
| $s \in \mathbb{R}^m$               | Measurement vector                                 |
| $A \in \mathbb{R}^{m \times n}$    | Sensing matrix                                     |
| $\Phi \in \mathbb{R}^{n \times K}$ | Learned PCA latent basis                           |
| $\alpha \in \mathbb{R}^K$          | Latent coefficient vector                          |
| $K$                                | Number of retained latent components               |
| $m_{\text{LED}}$                   | Number of LED channels added to RGB                |
| $F_\alpha$                         | Fisher information matrix in latent space          |
| $\kappa(A\Phi)$                    | Condition number of the effective sensing operator |
| $\Delta E_{00}$                    | CIEDE2000 color difference [5,16]                  |

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